

Math 526, S16

Quiz 3

Section:

Name: Aswer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Given a system of equations

$$x + 2y + 3z = 1$$

$$2x + 4y - z = 4$$

$$3y + z = 3$$

(a) Write down the augmented matrix $[A, b]$.

$$[A, b] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 4 & -1 & 4 \\ 0 & 3 & 1 & 3 \end{array} \right]$$

(b) Find the *Elimination matrix* E_{21} which can be used to eliminate the first component below the first pivot. Calculate $E_{21}[A, b]$.

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_{21}[A, b] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 0 & -7 & 2 \\ 0 & 3 & 1 & 3 \end{array} \right]$$

(c) Find the *Permutation matrix* P_{23} such that $P_{23}E_{21}[A, b]$ becomes triangular.

$$P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P_{23}E_{21}[A, b] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 3 & 1 & 3 \\ 0 & 0 & -7 & 2 \end{array} \right]$$

Problem 2. (4 points) Let

$$A = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

(a) Compute A^2 and A^3 .

$$A^2 = \begin{bmatrix} 4 & 0 \\ 3 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 8 & 0 \\ 7 & 1 \end{bmatrix}$$

(b) Predict A^n based on results from (a).

$$A^n = \begin{bmatrix} 2^n & 0 \\ 2^n - 1 & 1 \end{bmatrix}$$